

Understanding Logs & Log Analysis Basics

TechBiz Security SOC Analyst Internship

Task Report

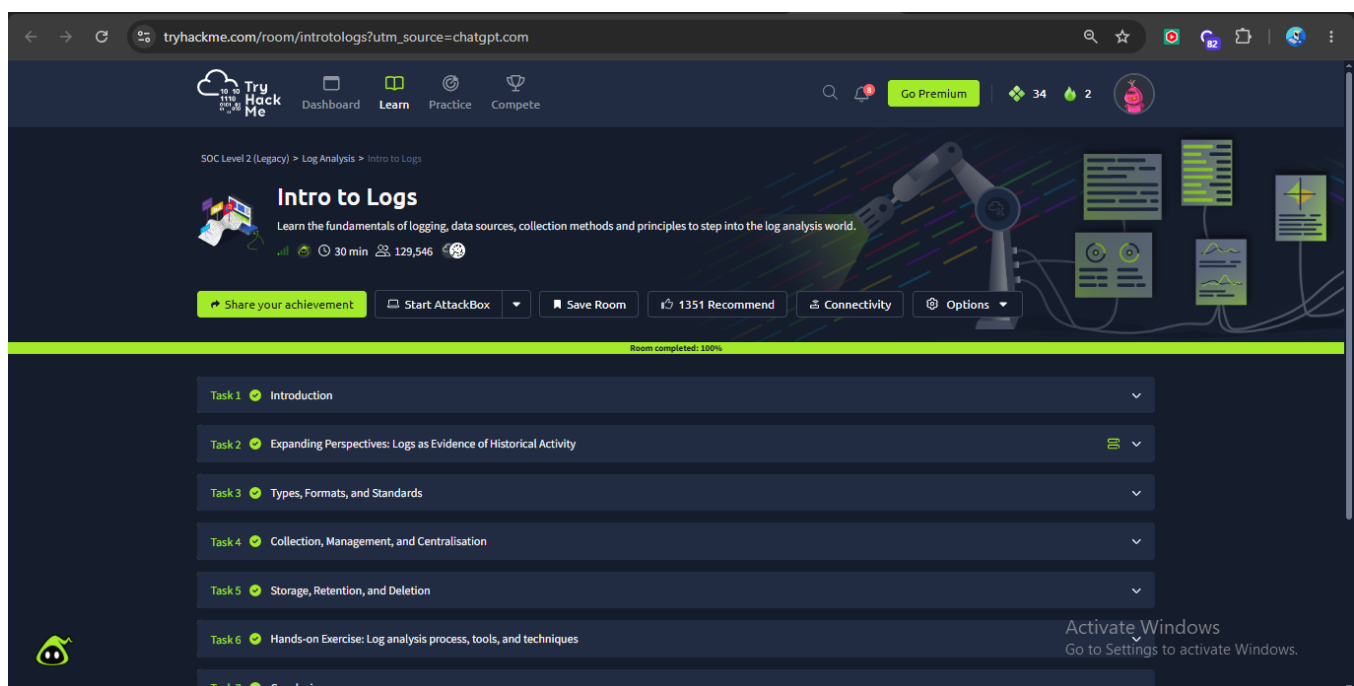
Project Overview

This report documents practical learning in log analysis, including common log sources, Apache web-server log interpretation, normal versus suspicious activity, pattern and anomaly analysis, and the importance of establishing a baseline.

TryHackMe Completion Proof

Intro to Logs

The Intro to Logs room was completed successfully, covering logging fundamentals, log types, formats, collection, storage, retention, and analysis.



Investigating Windows

The Investigating Windows room was completed successfully, providing practical exposure to investigating Windows activity and evidence available through Windows systems and logs.

tryhackme.com/room/investigatingwindows?utm_source=chatgpt.com

TryHackMe

DashboardLearnPracticeCompete

Go Premium

342

Learn > Investigating Windows



Investigating Windows

A windows machine has been hacked, its your job to go investigate this windows machine and find clues to what the hacker might have done.

45 min

149,627

Share your achievement

Start AttackBox

Save Room

2925 Recommendations

Connectivity

Options

Room completed: 100%

Task 1 Investigating Windows

How likely are you to recommend this room to others?

1

2

3

4

5

6

7

8

9

10

Submit now



Activate Windows

Go to Settings to activate Windows.

Five Common Log Types and Their Sources

Log Type	Source	What It Records	SOC Analyst Use
System Logs	Linux/Windows operating systems	System events, services, errors, startup/shutdown	Detect failures and unusual system activity
Security Logs	Windows Event Viewer, Linux <code>auth.log</code> , <code>secure</code>	Logins, failed logins, security changes, authentication events	Identify brute force and unauthorized access
Application Logs	Applications and software	Errors, user activity, transactions, application events	Identify application failures and suspicious behavior
Network Logs	Firewalls, routers, IDS/IPS, monitoring tools	Connections, IPs, ports, protocols, traffic	Detect malicious connections and unusual traffic
Web Server Logs	Apache and Nginx	HTTP requests, client IPs, URLs, status codes	Detect scanning, attacks, and suspicious requests

Practical Observation

A SOC analyst gains stronger context by correlating information from multiple log sources. For example, repeated failed logins combined with repeated network connections from the same source may indicate a brute-force attempt rather than an isolated user mistake.

Apache Web-Server Log Analysis

```
192.168.1.50 - - [30/Aug/2026:10:15:32 +0500] "GET /index.html HTTP/1.1" 200 4523
```

Field	Value	Meaning
Client IP	192.168.1.50	Device making the request
Timestamp	30/Aug/2026:10:15:32 +0500	Date, time, and timezone
HTTP Method	GET	Client requested information
Requested Resource	/index.html	Webpage/resource requested
Protocol	HTTP/1.1	HTTP protocol version
Status Code	200	Request processed successfully
Response Size	4523	Bytes returned by the server

Analysis

This entry appears normal because a legitimate webpage was requested and the server returned an HTTP 200 OK response. No obvious malicious path or suspicious repetition is visible.

Normal vs Suspicious Log Entries

Normal Activity

```
192.168.1.50 - - [30/Aug/2026:10:15:32 +0500] "GET /index.html HTTP/1.1" 200 4523
```

Classification: Normal Activity. The request targets a common webpage, uses a standard GET request, and succeeds without an obvious suspicious pattern.

Suspicious Activity

```
203.0.113.25 ... "GET /admin HTTP/1.1" 404 512
203.0.113.25 ... "GET /administrator HTTP/1.1" 404 512
203.0.113.25 ... "GET /wp-admin HTTP/1.1" 404 512
203.0.113.25 ... "GET /.env HTTP/1.1" 404 512
```

Classification: Suspicious Activity — Possible Web Enumeration or Automated Scanning. The same source rapidly requests multiple sensitive or commonly targeted paths. One 404 can be normal, but this sequence forms a suspicious pattern.

Pattern and Anomaly Analysis

Frequency

One failed login may be a normal mistake; hundreds of failed logins may indicate a brute-force attack.

Timing

Requests spread over hours may be normal, while many requests within seconds can indicate automation.

Source

A source browsing normal pages may be expected; a source repeatedly probing administrative paths requires investigation.

Behavior

Normal browsing may follow `/index.html` → `/products` → `/contact`. Reconnaissance may follow `/admin` → `/wp-admin` → `/.env` → `/phpmyadmin`.

Understanding a Baseline

A baseline is an established understanding of normal activity for a system, application, user, or network. Analysts use this expected behavior to identify deviations that may require investigation.

Example baseline: 500–700 website requests per hour, mostly during business hours, frequent requests for normal pages, and few failed requests.

Example anomaly: 10,000 requests in five minutes, repeated requests for `/admin`, many 404 responses for sensitive files, or one source generating thousands of requests.

SOC Analyst Workflow

Collect Logs → Understand Normal Behavior → Create a Baseline → Monitor New Events → Identify Deviations → Investigate Patterns → Determine Whether Activity Is Suspicious

Practical Learning Completed

- Identified five major log types and their sources.
- Read and interpreted common log fields including timestamps, IP addresses, requests, and status codes.
- Broke down an Apache web-server access-log entry.
- Compared normal and suspicious log activity.
- Identified patterns based on frequency, timing, source, and behavior.
- Understood how baselines support anomaly detection.

Conclusion

Logs are a primary source of evidence for SOC analysts. Effective analysis requires more than reading isolated events: an analyst must understand the source, interpret the fields, establish a baseline of normal behavior, and investigate patterns that deviate from that baseline.